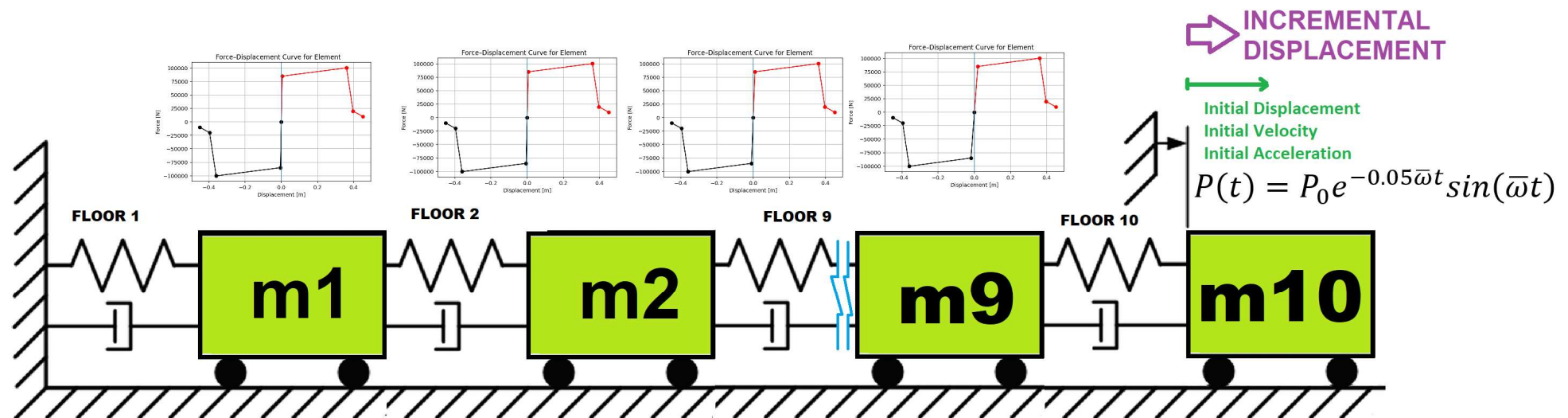


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF THE OPTIMUM ELASTIC STIFFNESS OF EACH COLUMN BASED ON THE CONSTANT STRUCTURAL PERIOD THROUGH MASS VARIATION USING FINITE- DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

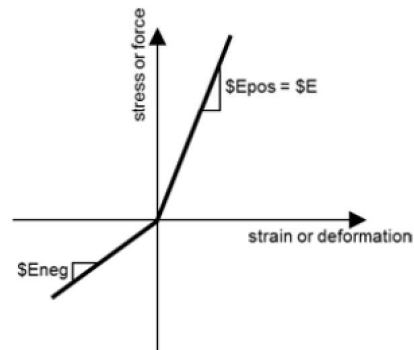
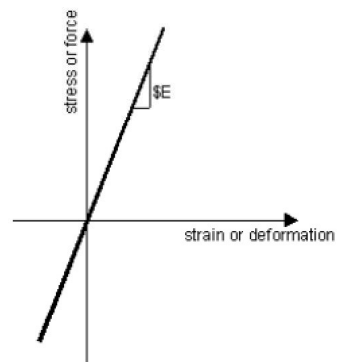
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

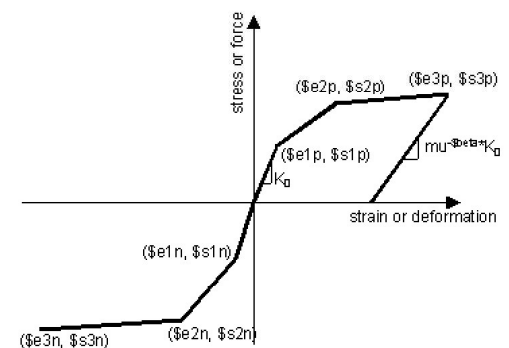
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

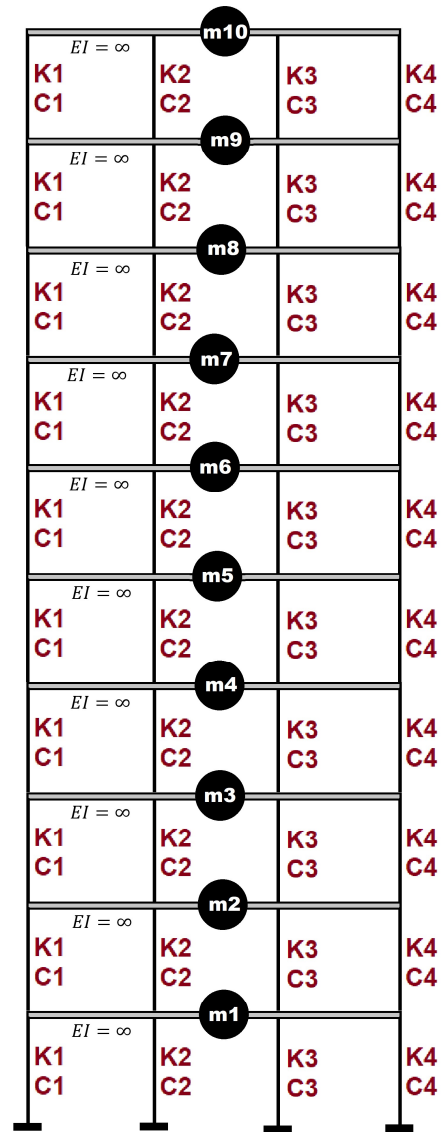
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



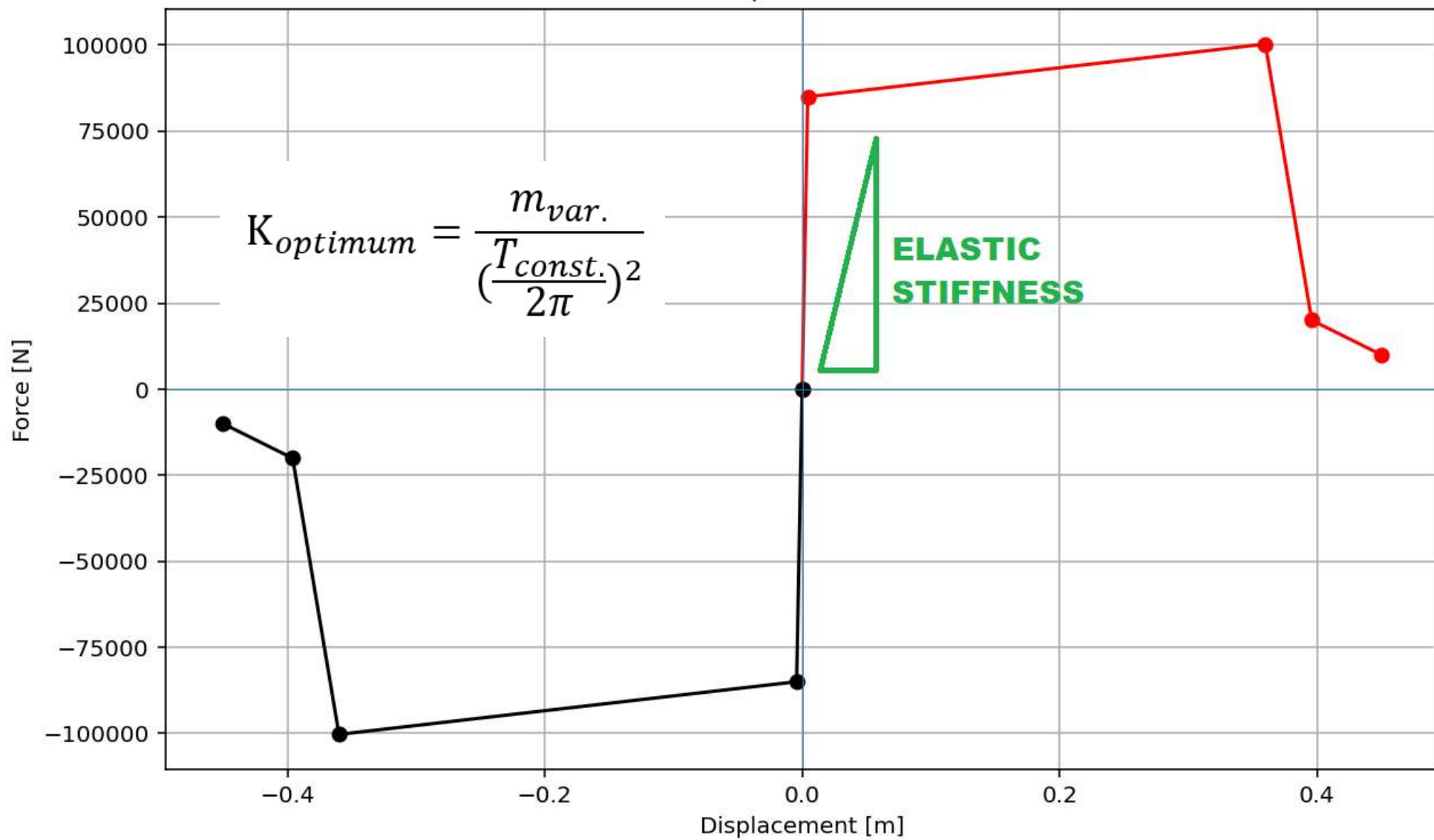
ELASTIC MATERIAL



INELASTIC MATERIAL



Force-Displacement Curve



**OPTIMIZATION
PROBLEM FOR
STRUCTURAL PERIOD
BY CHANGING INITIAL
ELASTIC STIFFNESS**

C:\Users\De\l\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS.py

KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py X PARA_KE_OPTIMIZATI...OORS_04_COLUMNS.py X

```

1201     print('DF: ', DF)
1202     # DX -----
1203     DX = F / DF; # Calculate dx
1204     print('DX: ', DX)
1205     # RESIDUAL -----
1206     RESIDUAL = np.abs(DX); # Calculate residual
1207     print('IT: ', IT + 1, ' - RESIDUAL: ', RESIDUAL, 'X: ', X, '\n')
1208     X -= DX; # update X
1209     IT += 1; # update iteration
1210     # CONTROLLING -----
1211     if IT == ITMAX:
1212         print("\t\t Iteration reached to Max. Iteration")
1213         print("\t\t Change ESP and TOLERANCE for better Convergence")
1214         X -= DX # update X
1215         break;
1216     if RESIDUAL < TOLERANCE:
1217         print(f'\t\t Structural Mass: {TOTAL_MASS:.4f}')
1218         print(f'\t\t Optimum X (COL. ELASTIC STIFFNESS): {X:.4f}')
1219         print(f'\t\t Iteration Counts: {IT}')
1220         print(f'\t\t Convergence Residual: {RESIDUAL:.10e}')
1221     #print(X)
1222
1223     OPT_COL_KE_target.append(X)
1224     TOTAL_MASS_target.append(TOTAL_MASS)
1225
1226     totaltime = TI.process_time() - starttime
1227     print(f'\nTotal time (s): {totaltime:.4f} \n\n')
1228
1229
1230     XDATA = OPT_COL_KE_target
1231     YDATA = TOTAL_MASS_target
1232     XLABEL = 'OPTIMUM ELASTIC COL. STIFFNESS [N/m]'
1233     YLABEL = 'STRUCTURAL TOTAL MASS [kg]'
1234     TITLE = f'{YLABEL} and {XLABEL} DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS \n TARGETED STRUCTURAL PI

```

..._8_ANA\OPTIMIZATION_40_PERIOD_MDOF_10_FLOORS_04_COLUMNS+

Console 1/A X

1030 2030
1037 2037
1038 2038
1039 2039

lambda	omega	period	frequency
1.579e+02	12.5664	0.5000	2.0000
1.395e+03	37.3563	0.1682	5.9454
3.737e+03	61.1273	0.1028	9.7287
6.927e+03	83.2309	0.0755	13.2466
1.062e+04	103.0642	0.0610	16.4032

Fmax: -6.693534615465069e-13
 DF: -6.682432385218817e-10
 DX: -6.645622196378136e-07
 IT: 320 - RESIDUAL: 6.645622196378136e-07 X: 373495321.34331524

 Structural Mass: 2000000.0000
 Optimum X (COL. ELASTIC STIFFNESS): 373495321.3433
 Iteration Counts: 320
 Convergence Residual: 6.6456221964e-07

 Total time (s): 3.8125

STRUCTURAL TOTAL MASS [kg] and OP

Fmax: -6.693534615465069e-13

DF: -6.682432385218817e-10

DX: -6.645622196378136e-07

IT: 320 - RESIDUAL: 6.645622196378136e-07 X: 373495321.34331524

Structural Mass: 2000000.0000

Optimum X (COL. ELASTIC STIFFNESS): 373495321.3433

Iteration Counts: 320

Convergence Residual: 6.6456221964e-07

Total time (s): 3.8125

STRUCTURAL TOTAL MASS [kg] and OPTIMUM ELASTIC COL. STIFFNESS [N/m] DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED STRUCTURAL PERIOD: 0.500 [s]

